O LEVEL PROGRAMMING PRACTICE QUESTIONS

DUE DATE: 06 NOVEMBER 2026

1. Core programming practice

Attempt these first. They strengthen the logic you will later place inside Windows Forms event procedures.

Question 1: Input, variables and basic calculations (10 marks)

A stationery shop sells exercise books for $1.50 each and pens for $0.80 each. Write a program that accepts the number of exercise books and pens purchased.

(a) Declare suitable variables and constants for the quantities and prices.
(b) Calculate the cost of exercise books and the cost of pens.
(c) Calculate and display the total amount payable.
(d) Use meaningful variable names and suitable data types.

Question 2: Rectangle calculator (10 marks)

Write a program that accepts the length and width of a rectangle.

(a) Calculate the area and perimeter.
(b) Display whether the rectangle is a square or not.
(c) Reject zero or negative dimensions.
(d) Display clear output labels.

Question 3: Examination grade checker (12 marks)

A school awards grades according to the following scale: A = 80–100, B = 70–79, C = 60–69, D = 50–59 and F = 0–49.

(a) Accept a learner name and examination mark.
(b) Validate that the mark is between 0 and 100.
(c) Use selection statements to determine the grade.
(d) Display the learner name, mark, grade and Pass/Fail status. A pass is 50% or above.
(e) Ensure that the boundaries 49/50, 59/60, 69/70 and 79/80 are handled correctly.

Question 4: Simple interest calculator (10 marks)

A savings group calculates simple interest using SI = (P × R × T) / 100.

(a) Accept principal, annual interest rate and time in years.
(b) Calculate simple interest.
(c) Calculate the final amount (principal plus interest).
(d) Display all inputs and results with suitable messages.
(e) Validate that the principal, rate and time are not negative.

Question 5: Repetition and a multiplication table (10 marks)

Write a program that accepts a whole number and displays its multiplication table from 1 to 12.

(a) Use an appropriate loop.
(b) Display each multiplication in the form number × multiplier = answer.
(c) Reject invalid input.
(d) Add a loop that allows the user to generate another table until they choose to stop.

Question 6: Functions and procedures (15 marks)

Write a modular program that accepts the marks for four subjects for one learner.

(a) Create a function **CalculateTotal** that accepts the four marks and returns the total.
(b) Create a function **CalculateAverage** that accepts the total and returns the average.
(c) Create a function **DetermineGrade** that accepts the average and returns a grade.
(d) Create a procedure **DisplayReport** that displays the learner name, marks, total, average and grade.
(e) Call the functions and procedure from the main program or the appropriate Windows Forms event procedure.
Extension: Add a function that returns True when the learner has passed and False otherwise.

Question 7: Testing and debugging (12 marks)

A program accepts two numbers and calculates their average. Learners have reported that it sometimes crashes and sometimes gives the wrong answer.

(a) State the difference between a syntax error, a runtime error and a logic error.
(b) Give one example of each type of error in this program.
(c) Identify suitable normal, boundary and invalid test data.
(d) Explain how a breakpoint or step-by-step execution can help locate an error.
(e) Modify the program so that invalid numeric input is handled safely.